

Manikanta Tanniru

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SUMMARY

Software Engineer with **5+ years building high-scale backend and AI-powered platforms**, specializing in **production LLM integration, workflow automation, and cloud-native system design**. Skilled in Java, Python, TypeScript, React distributed systems, and REST APIs, with proven ability to translate complex business problems into measurable efficiency, cost, and productivity gains.

SKILLS

Languages: Java, Python, Scala, TypeScript, JavaScript, C#/C

Cloud & DevOps: AWS (EKS, EMR, Lambda, DynamoDB, S3, RDS, IAM, CloudWatch), Azure (Databricks, Data Factory), Terraform, Docker, Kubernetes

Data & Streaming: Apache Spark, PySpark, Scala Spark, Kafka, Pulsar, Databricks, Performance Tuning

Databases: MS SQL Server, Oracle, SSMS, SSIS, DynamoDB

Web & APIs: React.js, Angular, Node.js, HTML5, CSS3, RESTful APIs

Generative AI & ML: Amazon Bedrock (Claude, Titan), Agent-based Workflows, Prompt Engineering, Human-in-the-Loop AI, Model Guardrails

Tools & CI/CD: Selenium, Git, GitHub Actions, Jenkins, IntelliJ, Visual Studio, TestNG, JUnit

EXPERIENCE

VeroTX Networks, San Jose, CA | AI Software Engineer

May 2026 – Present

- Designed, built, and operated large-scale backend services and **Implemented agent-orchestrated, multi-step AI workflows** (planning, generation, evaluation, approval) with throttling, retries, and failure isolation to safely handle hundreds of concurrent AI requests.
- Architected low-latency, highly available serverless and microservices systems using AWS Lambda, DynamoDB, SQS, and Step Functions, maintaining P99 latency under 200 ms during peak traffic.
- Built the front-end for a user-facing access portal in React/TypeScript, implementing component architecture, state management, and responsive UI to consolidate team, access, and status views into a single interface
- Built and operationalized production AI and LLM-based platforms using Amazon Bedrock to automate personalized content generation, reducing manual content creation effort by 68% and accelerating campaign delivery.
- Designed secure, versioned data storage using DynamoDB to persist millions of AI-generated artifacts, approvals, and audit records with IAM-based access controls and auditability.
- Strengthened system resilience by implementing timeouts, retries, circuit breakers and graceful fallbacks, reducing customer-impacting failures by 30%.
- Built high-throughput REST APIs using Node.js and TypeScript, supporting thousands of requests per minute with robust validation, idempotency, and error handling.
- Established end-to-end observability using CloudWatch, structured logging, dashboards, and alerts, reducing mean time to detect (MTTD) incidents by 40% and improving on-call signal quality.

Amazon, Sunnyvale, CA | SDE II

Feb 2025 – May 2026

- Involved in various phases of (SDLC) as requirement gathering, modelling, analysis, architecture design & development and the project was developed using Agile Methodologies.
- Worked on Hummingbird, an event-driven integration platform enabling real-time data exchange across Amazon Robotics systems and external partners.
- Designed and implemented workflow-based integrations between systems like Oracle EBS, Kinaxis, PLM, SAP, and Proplanner. Built AWS-based orchestration pipelines using Step Functions, EventBridge, Lambda, and SQS for scalable.
- Built large-scale dataset transformation pipelines using PySpark and Spark SQL on Databricks, processing millions of records for downstream analytics and content workflows
- Contributed to self-service workflow onboarding, reducing onboarding time significantly. Supported high-scale system handling 100K+ requests/sec and 300+ API routes per tenant.
- Worked on the end-to-end architecture and implementation of Project Horus, an automated testing platform for Amazon Robotics, enabling business process workflow automation and reducing manual testing effort by over 3000 hours annually.
- Designed and helped implement a new user-facing portal on top of the existing Mockingbird access automation service, giving Amazon Robotics users a single place to view their teams, app access, and status instead of relying on scattered wiki pages. Optimized backend flows between the Self-Service Lambda, Connector Lambdas, DeprovisionService.

DarioHealth, USA | Full Stack Developer

June 2023 – Feb 2025

- Developed backend and full-stack applications with Java, Node.js, and REST APIs for enterprise marketing and analytics platforms in a client delivery environment.
- Built backend services for automated campaign and content workflows, reducing manual campaign setup effort by 40% for marketing operations teams.
- Built AWS Lambda/Step Functions automation for multi-stage content generation, validation, and approval, cutting end-to-end turnaround from days to hour
- Integrated early LLM-based content generation for cohort-specific marketing, reducing repetitive manual writing by 60–65%; added rule-based quality checks that cut rework by 30%.

Veritas Software, India | Software Developer

June 2019 – July 2021

- Improved code quality using Java 8 Stream API, Lambda expressions, and Concurrency API; built Python (Django/Flask) REST APIs with auth and DB integration, raising reliability by 20%.
- Designed data persistence and versioning with DynamoDB and relational databases to track content iterations, approvals, and audit history.

EDUCATION

- Doctor of Business Administration (DBA), Business Intelligence & Data Analytics — Westcliff University, San Francisco, CA (June 2026 – 2029)
- M.S. Software Engineering — San José State University, San Jose, CA (May 2023)
- B.E. Computer Science & Engineering — K L University, Andhra Pradesh (May 2019)

KEY ACHIEVEMENTS

- Delivered AI-powered automation platforms integrating LLMs into production, reducing manual operational effort by 60–90% across marketing and content workflows.
- Designed and scaled cloud-native AWS systems (portable across cloud platforms) handling millions of monthly requests with low latency and high availability.
- Drove \$70M+ in measurable business impact through platform modernization, workflow automation, and performance optimization initiatives.